

Excercise 1 Basics of Junit

Code Unit

```
Calculator.java x CalculatorTest.java
package org.example;

public class Calculator { 2 usages

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) { 1 usage
        return a - b;
    }

    public int multiply(int a, int b) { 1 usage
        return a * b;
    }

    public int divide(int a, int b) { 1 usage
        return a / b;
    }
}
```

Dependencies

```
<dependencies>

    <dependency>
        <groupId>junit</groupId>
        <artifactId>junit</artifactId>
        <version>4.13.2</version>
        <scope>test</scope>
    </dependency>

</dependencies>
```

Test Code

```
3  import org.junit.Test;
4  import static org.junit.Assert.*;
5
6  public class CalculatorTest {
7
8      Calculator calculator = new Calculator(); 4 usages
9
10     @Test
11     public void testAddition() {
12
13         assertEquals("expected: 10, calculator.add(a: 5, b: 5));", 10, calculator.add(5, 5));
14     }
15
16
17     @Test
18     public void testSubtraction() {
19
20         assertEquals("expected: 4, calculator.subtract(a: 9, b: 5));", 4, calculator.subtract(9, 5));
21     }
22
23
24     @Test
25     public void testMultiplication() {...}
26
27
28     @Test
29     public void testDivision() {
30
31         assertEquals("expected: 5, calculator.divide(a: 25, b: 5);", 5, calculator.divide(25, 5));
32     }
33
34
35
36 }
```

```
Run CalculatorTest x
[Icons]
v ✓ CalculatorTest (org.example) 4 ms ✓ 4 tests passed 4 tests total, 4 ms
  ✓ testMultiplication 4 ms "C:\Program Files\Java\jdk-21\bin\java.exe" ...
  ✓ testAddition 0 ms Process finished with exit code 0
  ✓ testDivision 0 ms
  ✓ testSubtraction 0 ms
Calculator > src > test > java > org > example > CalculatorTest > testSubtraction
```